

- 21) Which of the following instructions converts any BCD value available in AX register into ASCII equivalent of a BCD code can be obtained by appending $(011)_2$ to left of the BCD value.
- A) AND AX, 07h
B) OR AX, 0Fh
C) AND AX, 0Fh
D) XOR AX, 0Fh
E) OR AX, 30h
- 22) Assume that SP=0100H. What is the value of SP after execution of POP AX instruction?
- A) 00FFH B) 0101H C) 0102H D) 00FEH
- 23) Which of the following communication architectures are parallel?
- I. USB
II. PCI
III. PCI Express
IV. ISA
- A) I and II B) I, II, III C) I and III D) II and IV
- 24) How many (128Mx16) memory device is needed to store 1 G bytes of data?
- A) 1 B) 16 C) 2 D) 8
- 25) Which of the following connections are common to all memory devices?
- I. Address
II. Read Control
III. Write Control
IV. Data
- A) I, II, IV B) I, II, III, IV C) Only IV D) I, II, III, IV

- 2) Which type of jump instruction is used to jump to any location?
- I. Short
II. Near
III. Far
IV. Relative
- A) I and II B) I and III C) II and IV D) Only III E) I and IV
- 3) How many (128Mx16) memory device is needed to store 1 G bytes of data?
- A) 2 B) 8 C) 4 D) 16 E) 1
- 4) INC instruction is a special type of addition. However, _____ flag is not affected by INC instruction unlike ADD.
- A) S B) Z C) C D) P E) A
- 5) Regarding SRAM and DRAM, which of the following statements are true?
- I. DRAM is faster than SRAM.
II. SRAM is more expensive than DRAM.
III. DRAM requires periodical refresh.
IV. SRAM consumes more power than DRAM.
- A) Only I B) III and IV C) I, II, IV D) I and IV E) II and III
- 6) Which of the following flag bits is not tested by the conditional jump instructions?
- A) S B) C C) D D) O E) P
- 7) TEST instruction performs _____ operation that affects the flags register.
- A) OR B) NOR C) XOR D) NOT E) AND
- 8) Which of the following flags indicates whether the result of an arithmetic operation is positive or negative?
- A) S B) C C) O D) Z E) D

1) Instruction Queue is placed in _____ of Intel 8086.

- A) Control and Timing Unit
- B) Execution Unit
- C) Register File
- D) Bus Interface Unit
- E) Arithmetic Logic Unit

2) _____ is the act of synchronizing two systems that op

- A) Serial communication
- B) Handshaking
- C) Transmission
- D) Parallel communication
- E) Peripheral interface

3) Which of the followings are interrupt pins found on Intel micro

- I. INTA
- II. INTC
- III. INTQ
- IV. INTR

- A) I and IV
- B) I, III, IV
- C) Only III

16) In 8086, 32-bit addition can be performed by adding low 16-bits of the operand and high 16-bits of the operands using _____ instruction.

- A) ADC - SBB
- B) ADD - ADC
- C) ADC - ADC
- D) ADD - ADD
- E) ADC - ADD

17) What is the IEEE 754 floating point equivalent of $-\infty$?

- A) 10000000011111111111111111111111b
- B) 10000000000000000000000000000000b
- C) 00000000000000000000000000000000b
- D) 11111111100000000000000000000000b
- E) 11111111111111111111111111111111b

18) It is given that the following code computes factorial of var_N. Then, what is the missing instruction?

```
MOV AX, 1
MOV BX, var_N
MOV CX, var_N
label:
MUL BX
missing instruction?
LOOP label
RET
var_N DW 6
```

- A) INC AX
- B) DEC CX
- C) INC BX

19) Which field of an instruction specifies the addressing mode to be used?

- A) REG
- B) R/M
- C) Direction

20) Assume that CS=1000H, DS=2000H, SS=3000H, and IP=2000H. Then, what is the address of the next instruction executed by the microprocessor?

- A) 12000H
- B) 22000H
- C) 1000H

4) Which of the following communication architectures are serial?

- I. USB
- II. PCI
- III. PCI Express
- IV. ISA

- A) I, II, IV B) I, II, III C) I and III D) Only I

5) Which flag is tested by a conditional interrupt instruction?

- A) Interrupt B) Carry C) Parity D) Overflow

6) Regarding SRAM and DRAM, which of the following statements are true?

- I. DRAM is faster than SRAM.
- II. SRAM is more expensive than DRAM.
- III. DRAM requires periodical refresh.
- IV. SRAM consumes more power than DRAM.

- A) II and III B) I, II, IV C) Only I D) I and IV

7) What is the IEEE 754 single precision floating point representation of decimal +10.07

- A) 41200000h B) C1A00000h C) 41A00000h D) C1200000h

8) Which of the following statements is not true when an interrupt occurs?

- I. Flags register contents are pushed on the stack.
- II. Interrupt flag is set.
- III. Contents of CS and IP are pushed onto the stack.
- IV. Related ISR is executed.

- A) Only II B) I and III C) III and IV D) II and IV

9) Using little endian format, 4321h is to be stored in an 8-bit memory starting at 20h memory address 201h after storing?

- A) 34h B) 43h C) 12h D) Empty

8) Which of the following memory types are volatile?

- I. SRAM
- II. DRAM
- III. Flash
- IV. EPROM

- A) I, II, IV B) III and IV C) I and II D) Only IV E) I, II, III, IV

9) A short jump allows a program to branch forward _____ bytes or backwards _____ bytes from the next instruction's address in the program.

- A) 127, 128 B) 128, 127 C) 256, 255 D) 255, 256 E) 32K, 32K

10) 12-bit Hamming coded data including 8-bit actual data and 4-bit parity is received as given below.

Bit	1	2	3	4	5	6	7	8	9	10	11	12
Data	0	0	1	1	1	0	0	1	0	0	0	0

Using the error checking method provided below, detect position of the error (if any) within the code.

C1 = XOR of bits (1, 3, 5, 7, 9, 11)

C2 = XOR of bits (2, 3, 6, 7, 10, 11)

C4 = XOR of bits (4, 5, 6, 7, 12)

C8 = XOR of bits (8, 9, 10, 11, 12)

A) Bit-2

B) Bit-6

C) No error

D) Bit-10

E) Bit-4

Using the error checking method provided below, detect position of the error

C1 = XOR of bits (1, 3, 5, 7, 9, 11)

C2 = XOR of bits (2, 3, 6, 7, 10, 11)

C4 = XOR of bits (4, 5, 6, 7, 12)

C8 = XOR of bits (8, 9, 10, 11, 12)

A) Bit-2

B) Bit-4

C) Bit-6

D)

11) What is the corresponding value of -76.5 in nearest even rounding mode?

A) -77

B) -75

C) -78

D)

12) The code below computes the sum of numbers from 1 to N using the for instruction should be inserted into the missing line within the code?

$$\sum_{x=1}^N x = N(N+1)/2$$

ORG 100h

CALL sum

RET

sum PROC

MOV AL, N

MOV BL, N

INC BL

MUL BL

missing instruction?

RET

sum ENDP

N DB 8

A) DEC AL

B) SHR AL, 1

C) SUB AL, 2

13) What is the decimal equivalent of IEEE 754 single precision floating point number?

A) -20.25

B) -40.125

C) -10.50

14) What is the size of Personal Computer I/O map?